# Find the Error

A.SSE.1

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In the expression 5x + 12, what is the constant term? Student's answer: 5x

**Identify the part of the expression with no variable.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **5x has the variable, so it is not constant. The constant term is 12.** | **12** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In the expression 7 - 2t, what is the coefficient of t? Student's answer: 2

**Include the sign with the coefficient.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The coefficient is the number multiplied by t, including its sign. It is -2.** | **-2** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

n costs 25 + 10g dollars, where g is the number of GB used. What does 25 represent? Student's answer: the

**Decide which part does not change with usage.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The cost per GB is the 10. The 25 is the fixed part that is paid no matter how much data is used.** | **The fixed monthly fee** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ength l and width w. The expression 2(l + w) gives the perimeter. What does l + w represent? Student's answ

**Think about how the perimeter is written.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The perimeter is 2(l + w), so l + w is half the perimeter.** | **Half the perimeter** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ation is P = 800 × 0.9^t, where t is the number of years. What does 0.9 represent? Student's answer: 90% gro

**Decide whether the population grows or shrinks.**

What did the student do wrong? Why do you think the student made this error?

**Because the base is less than 1, the population shrinks. Each year it keeps 90%, a 10% decrease.**

Solve the problem correctly. Show all of your work.

**Each year it keeps 90%, so it decreases 10% per year**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What are the terms of $4x^2 - 3x + 9$? Student's answer: $4x^2$, 3x, 9

**Keep the sign with each term.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The terms include their signs. The term is −3x, not 3x.** | **$4x^2$, −3x, 9** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $x^2 - 9$. Student writes: $(x - 3)(x - 3)$

**Look for two numbers that multiply to −9 and add to 0.**

What did the student do wrong? Why do you think the student made this error?

**$(x - 3)(x - 3) = x^2 - 6x + 9$. The correct factorization is $(x - 3)(x + 3)$.**

Solve the problem correctly. Show all of your work.

**$(x - 3)(x + 3)$**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $x^2 + 5x + 6$. Student writes: $(x + 1)(x + 6)$

**Find two numbers that multiply to 6 and add to 5.**

What did the student do wrong? Why do you think the student made this error?

**1 + 6 = 7, not 5. The numbers are 2 and 3, so the answer is (x + 2)(x + 3).**

Solve the problem correctly. Show all of your work.

**(x + 2)(x + 3)**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $4x^2 - 25$ as a difference of squares. Student writes: $(4x - 5)(x + 5)$

**Write each term as a square.**

What did the student do wrong? Why do you think the student made this error?

**$4x^2 = (2x)^2$ and $25 = 5^2$. The factorization is $(2x - 5)(2x + 5)$.**

Solve the problem correctly. Show all of your work.

**$(2x - 5)(2x + 5)$**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $3x^2 + 6x$ completely. Student writes: $3(x^2 + 2x)$

**Look for every common factor.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student did not factor the remaining $x^2$. The GCF is $3x$, so the answer is $3x(x + 2)$.** | **$3x(x + 2)$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Use the difference of squares to find 51 × 49 without a calculator. Student's answer: 2,500

**Write 51 and 49 as 50 + 1 and 50 − 1.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **$(50 + 1)(50 - 1) = 50^2 - 1 = 2{,}500 - 1 = 2{,}499.$** | **2,499** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $x^2 - 16$. Student writes: $(x - 4)^2$

**Check by multiplying.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **$(x - 4)^2 = x^2 - 8x + 16$. The correct factorization is $(x + 4)(x - 4)$.** | **$(x + 4)(x - 4)$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(x + 3)(x + 5)$. Student writes: $x^2 + 15$

**Use FOIL and combine like terms.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student left out the middle terms. $(x + 3)(x + 5) = x^2 + 8x + 15$.** | **$x^2 + 8x + 15$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add $(2x - 1) + (3x + 4)$. Student writes: $5x + 5$

**Combine like terms.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The constants are $-1 + 4 = 3$, not 5. The sum is $5x + 3$.** | **$5x + 3$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(x + 2)(x - 2)$. Student writes: $x^2 + 4$

**Multiply each pair of terms.**

What did the student do wrong? Why do you think the student made this error?

**The outer and inner terms cancel, leaving $x^2 - 4$. The student made a sign error.**

Solve the problem correctly. Show all of your work.

**$x^2 - 4$**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract $(3x^2 + x) - (x^2 - 4x)$. Student writes: $2x^2 - 3x$

**Distribute the negative sign to both terms of the second polynomial.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student did not change the sign of $-4x$. The answer is $2x^2 + 5x$.** | **$2x^2 + 5x$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Expand $(x - 4)^2$. Student writes: $x^2 + 16$

**Write the square as a product of two binomials.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The middle term is missing. $(x - 4)(x - 4) = x^2 - 8x + 16$.** | **$x^2 - 8x + 16$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(2x + 1)(x - 3)$. Student writes: $2x^2 - 6x - 3$

**Multiply each pair of terms, then combine.**

What did the student do wrong? Why do you think the student made this error?

**The student did not combine the x terms. $2x(-3) = -6x$ and $1(x) = x$, so the answer is $2x^2 - 5x - 3$.**

Solve the problem correctly. Show all of your work.

**$2x^2 - 5x - 3$**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an equation: a number increased by 7 is 19. Student writes: x - 7 = 19

**Translate the words into operations.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Increased by means add. The equation is x + 7 = 19.** | **x + 7 = 19** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A taxi charges $3 plus $2 per mile. The total fare is $15. Write an equation. Student writes: $2(m + 3) = 15$

**Decide which part is multiplied by the miles.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Only the miles are multiplied by $2. The $3 is added once. The equation is $3 + 2m = 15$, so $m = 6$.** | **$3 + 2m = 15$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

bakery sells 12 cookies per day. Write an inequality for at least 120 cookies in d days. Student writes: 12d ≤ 12

**Think about the words at least.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **At least means greater than or equal to. The inequality is 12d ≥ 120.** | **12d ≥ 120** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an equation: twice a number minus 4 is 18. Student writes: 2x + 4 = 18

**Translate each phrase in order.**

What did the student do wrong? Why do you think the student made this error?

**The phrase is minus 4, not plus 4. The equation is 2x - 4 = 18.**

Solve the problem correctly. Show all of your work.

**2x - 4 = 18**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ctangle has a length 3 more than its width w. The perimeter is 30. Write an equation. Student writes: 2w + 3

**Use the perimeter formula with both lengths and widths.**

What did the student do wrong? Why do you think the student made this error?

**The student forgot the length and the second width. The perimeter is 2(w + (w + 3)) = 30.**

Solve the problem correctly. Show all of your work.

**2(w + w + 3) = 30**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pens cost p dollars each, plus a $5 bag. The total is at most $25. Write an inequality. Student writes: 4p + 5

**Think about the words at most.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **At most means less than or equal to. The inequality is 4p + 5 ≤ 25.** | **4p + 5 ≤ 25** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the point (2, 5) on the graph of $y = 2x + 1$? Student says: No

**Substitute 2 for x.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **2(2) + 1 = 5, so the point is on the graph.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line passes through (0, 3) and (1, 5). Write its equation. Student writes: y = 3x + 2

**Find the slope, then the y-intercept.**

What did the student do wrong? Why do you think the student made this error?

**The slope is 2 and the y-intercept is 3. The equation is y = 2x + 3.**

Solve the problem correctly. Show all of your work.

**y = 2x + 3**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the y-intercept of y = -x + 4? Student's answer: -1

**Find the value of y when x = 0.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student used the slope, -1, instead of the y-intercept. When x = 0, y = 4.** | **4** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A car travels at 55 miles per hour. Let x be hours and y be miles. Write an equation. Student writes: y = x/55

**Decide whether miles depend on hours.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Distance grows by 55 for each hour. The equation is y = 55x.** | **y = 55x** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the point (3, -2) on the graph of $y = -x + 1$? Student says: No

**Substitute 3 for x.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **-3 + 1 = -2, so the point is on the graph.** | **Yes** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which equation is parallel to y = 3x - 2? Student writes: y = -3x + 5

**Parallel lines have the same slope.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student used the opposite slope. Parallel lines have the same slope, so y = 3x + 5 is parallel.** | **y = 3x + 5** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

st $5. The total revenue is $200. Let a be the number of adult tickets and s the number of student tickets. W

**Multiply each count by its price.**

What did the student do wrong? Why do you think the student made this error?

**The money total uses prices. The equation is 10a + 5s = 200.**

Solve the problem correctly. Show all of your work.

**10a + 5s = 200**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e needs at least $500 by selling x items at $12 and y items at $8. Write an inequality. Student writes: 12x + 8y

**Think about the words at least.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **At least means greater than or equal to. The inequality is 12x + 8y ≥ 500.** | **12x + 8y ≥ 500** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A class has 30 students. There are 6 more girls g than boys b. Write the equations. Student writes: g = b − 6

**Write the relationship in words first.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **There are 6 more girls, so g = b + 6. The total is g + b = 30.** | **g + b = 30 and g = b + 6** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

osts $3 per meter, and you can spend no more than $240. Write an inequality for m meters. Student writes:

**Think about the words no more than.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **No more than means less than or equal to. The inequality is 3m ≤ 240.** | **3m ≤ 240** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$2 and pencils for $1. A student spends at most $20 and buys at least 4 pens. Let p be pens and q be pencil

**Write each condition as an inequality.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **At least 4 pens means p ≥ 4. The spending condition is 2p + q ≤ 20.** | **2p + q ≤ 20 and p ≥ 4** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

rden has an area of at most 60 square feet with length l and width w. Write an inequality. Student writes: lw

**Think about the words at most.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **At most means less than or equal to. The inequality is lw ≤ 60.** | **lw ≤ 60** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve A = lw for w. Student writes: w = Al

**Undo the multiplication by l.**

What did the student do wrong? Why do you think the student made this error?

**The student multiplied instead of dividing. Divide both sides by l: w = A/l.**

Solve the problem correctly. Show all of your work.

**w = A/l**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $P = 2l + 2w$ for w. Student writes: $w = (P + 2l)/2$

**Subtract 2l first, then divide by 2.**

What did the student do wrong? Why do you think the student made this error?

**The student added 2l instead of subtracting. The answer is $w = (P - 2l)/2$.**

Solve the problem correctly. Show all of your work.

**$w = (P - 2l)/2$**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve d = rt for t. Student writes: t = dr

**Undo the multiplication by r.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student multiplied by r. Divide both sides by r: t = d/r.** | **t = d/r** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $V = (1/3)Bh$ for h. Student writes: $h = V/(3B)$

**Multiply by 3, then divide by B.**

What did the student do wrong? Why do you think the student made this error?

**The student forgot the multiplication by 3 in the right order. Multiply both sides by 3 to get 3V = Bh, then h = 3V/B.**

Solve the problem correctly. Show all of your work.

**h = 3V/B**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $C = (5/9)(F - 32)$ for F. Student writes: $F = (9/5)C - 32$

**Multiply by 9/5, then undo the subtraction.**

What did the student do wrong? Why do you think the student made this error?

**The student subtracted 32 instead of adding it. $F = (9/5)C + 32$.**

Solve the problem correctly. Show all of your work.

**$F = (9/5)C + 32$**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $y = mx + b$ for $x$. Student writes: $x = y - b/m$

**Subtract b, then divide by m.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student did not subtract b from all of y. The correct answer is $x = (y - b)/m$.** | **$x = (y - b)/m$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3x + 2 = 8$. A student's first step is: $3x = 10$. Is this step correct? Student says: Yes

**Check the first step.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **To isolate 3x, subtract 2 from both sides: 3x = 6. Adding 2 was the mistake.** | **No, it should be 3x = 6** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lain why adding 2 to both sides of x - 2 = 5 keeps the solution the same. Student says: It changes the soluti

**Think about what the equal sign means.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Doing the same operation to both sides of an equation keeps the two sides equal, so the solution stays the same.** | **Both sides stay equal, so the solution stays the same** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $\sqrt{x} = -2$. Student's answer: $x = 4$

**Square both sides and check the result.**

What did the student do wrong? Why do you think the student made this error?

**A square root cannot be negative, so there is no solution. Squaring gives x = 4, but 4 does not work in the original equation.**

Solve the problem correctly. Show all of your work.

**No solution**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2(x - 1) = 8$. Student's answer: $x = 4.5$

**Distribute, then solve.**

What did the student do wrong? Why do you think the student made this error?

**The student did not multiply the -1 by 2. $2x - 2 = 8$ gives $x = 5$, or divide both sides by 2 first to get $x - 1 = 4$.**

Solve the problem correctly. Show all of your work.

**$x = 5$**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $5x = 3x + 6$. Student's answer: $x = 6$

**Subtract 3x from both sides.**

What did the student do wrong? Why do you think the student made this error?

**The student did not divide by the new coefficient. 2x = 6, so x = 3.**

Solve the problem correctly. Show all of your work.

**x = 3**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is multiplying both sides of an equation by 0 a valid step when solving? Student says: Yes

**Think about whether the solution is kept.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Multiplying by 0 makes both sides 0, which removes the information. It is not a valid step for solving.** | **No** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3x - 4 < 11$. Student writes: $x > 5$

**Add 4, then divide by 3.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Dividing by a positive number keeps the sign. $3x < 15$, so $x < 5$.** | **$x < 5$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $-2x + 3 \geq 9$. Student writes: $x \geq -3$

**Subtract 3, then divide by −2.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Dividing by a negative reverses the sign. $-2x \geq 6$ gives $x \leq -3$.** | **$x \leq -3$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2(x + 4) > 2x + 8$. Student says: All real numbers are solutions.

**Distribute, then simplify both sides.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Both sides simplify to $2x + 8$. The statement $2x + 8 > 2x + 8$ is never true, so there is no solution.** | **No solution** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $4x - 2 = 4x + 6$. Student's answer: $x = 2$

**Subtract 4x from both sides.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Subtracting 4x leaves -2 = 6, which is never true. There is no solution.** | **No solution** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x/4 - 1 \leq 2$. Student writes: $x \leq 3$

**Add 1, then multiply by 4.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student did not multiply by 4. $x/4 \leq 3$ gives $x \leq 12$.** | **$x \leq 12$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $-(x - 5) = 2x$. Student's answer: $x = -5/3$

**Distribute the negative sign, then collect the x terms.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **$-x + 5 = 2x$ gives $5 = 3x$, so $x = 5/3$. The student lost the sign.** | **$x = 5/3$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 = 49$. Student's answer: $x = 7$

**Take the square root of both sides.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Both 7 and -7 square to 49. The solutions are x = 7 and x = -7.** | **x = 7 or x = -7** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $(x - 3)^2 = 16$. Student's answer: $x = 7$

**Take the square root of both sides, including the negative root.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student found only one root. $x - 3 = 4$ or $x - 3 = -4$, so $x = 7$ or $x = -1$.** | **$x = 7$ or $x = -1$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + 6x + 9 = 0$. Student's answer: $x = 3$

**Factor the trinomial.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **$x^2 + 6x + 9 = (x + 3)^2$, so $x = -3$. The student changed the sign.** | **$x = -3$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 - 5x + 6 = 0$. Student's answer: $x = 1$ or $x = 6$

**Find two numbers that multiply to 6 and add to -5.**

What did the student do wrong? Why do you think the student made this error?

**The student mixed up sum and product. The numbers -2 and -3 work, so (x - 2)(x - 3) = 0 and x = 2 or x = 3.**

Solve the problem correctly. Show all of your work.

**x = 2 or x = 3**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + 4x = 5$ by completing the square. Student's answer: $x = -2 \pm \sqrt{5}$

**Add the square of half the coefficient to both sides.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student did not add 4 to both sides. $x^2 + 4x + 4 = 9$ gives $(x + 2)^2 = 9$, so $x = 1$ or $x = -5$.** | **$x = 1$ or $x = -5$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 - 2x - 8 = 0$ with the quadratic formula. Student's answer: $x = 2$ or $x = -4$

**Use $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student changed the sign of b incorrectly. Here a = 1, b = -2, c = -8, so x = (2 ± 6)/2, giving x = 4 or x = -2.** | **x = 4 or x = -2** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + y = 10$ and $2x - y = 8$. Student's answer: (4, 6)

**Add the equations to eliminate y.**

What did the student do wrong? Why do you think the student made this error?

**Adding gives $3x = 18$, so $x = 6$ and $y = 4$.**
**The student swapped the values.**

Solve the problem correctly. Show all of your work.

**(6, 4)**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $2x + 3y = 12$ and $x - y = 1$. Student's answer: (2, 3)

**Solve one equation for a variable and substitute.**

What did the student do wrong? Why do you think the student made this error?

**From $x = y + 1$, $2(y + 1) + 3y = 12$ gives $y = 2$ and $x = 3$. The student swapped the values.**

Solve the problem correctly. Show all of your work.

**(3, 2)**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + y = 5$ and $x + y = 7$. Student's answer: (1, 4)

**Compare the two equations.**

What did the student do wrong? Why do you think the student made this error?

**The same left side cannot equal both 5 and 7. The lines are parallel, so there is no solution.**

Solve the problem correctly. Show all of your work.

**No solution**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $3x - y = 4$ and $6x - 2y = 8$. Student's answer: $(2, 2)$

**Multiply the first equation by 2.**

What did the student do wrong? Why do you think the student made this error?

**The second equation is just twice the first. Both equations describe the same line, so there are infinitely many solutions.**

Solve the problem correctly. Show all of your work.

**Infinitely many solutions**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is (1, 2) a solution of the system $2x + y = 4$ and $x - y = -1$? Student says: No

**Substitute into both equations.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **2(1) + 2 = 4 and 1 - 2 = -1. The point works in both equations.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The system $4x + 2y = 10$ and $2x + y = 5$ is solved. Student says there is exactly one solution, $(2, 1)$.

**Multiply the second equation by 2 and compare.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Doubling $2x + y = 5$ gives $4x + 2y = 10$, which is the first equation. The system has infinitely many solutions.** | **Infinitely many solutions** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = 2x + 1$ and $y = -x + 7$. Student's answer: (1, 3)

**Set the two expressions equal.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **$2x + 1 = -x + 7$ gives $x = 2$ and $y = 5$. The student used $x = 1$ by mistake.** | **(2, 5)** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = 3x - 2$ and $y = 3x + 4$. Student's answer: (3, 7)

**Compare the slopes and intercepts.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Both lines have slope 3 but different intercepts, so they are parallel. There is no solution.** | **No solution** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $y = 2x$ and $x + y = 9$ by substitution. Student's answer: (6, 3)

**Substitute 2x for y.**

What did the student do wrong? Why do you think the student made this error?

**x + 2x = 9 gives x = 3 and y = 6. The student swapped the values.**

Solve the problem correctly. Show all of your work.

**(3, 6)**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $y = -2x + 8$ and $y = x - 1$. Student's answer: (2, 3)

**Set the two expressions equal and solve for x.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **$-2x + 8 = x - 1$ gives $x = 3$ and $y = 2$. The student swapped the values.** | **(3, 2)** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many solutions does the system $y = 4x + 1$ and $y = 4x - 3$ have? Student's answer: 1

**Compare the slopes and intercepts.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The lines have the same slope and different intercepts. They are parallel, so there are no solutions.** | **0** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + 2y = 8$ and $x - y = 2$. Student's answer: (2, 3)

**Subtract the second equation from the first.**

What did the student do wrong? Why do you think the student made this error?

**From $x = y + 2$, substitute: $(y + 2) + 2y = 8$ gives $y = 2$ and $x = 4$. The student's answer does not satisfy $x + 2y = 8$.**

Solve the problem correctly. Show all of your work.

**(4, 2)**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = 2x + 3$, what is $f(4)$? Student's answer: 8

**Substitute 4 for x, then calculate.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student forgot to add 3. $f(4) = 2(4) + 3 = 11$.** | **11** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table lists x = 1, 2, 2 with y = 3, 4, 5. Is this a function? Student says: Yes

**Check whether any input has two outputs.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The input 2 has outputs 4 and 5, so it is not a function.** | **No** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = x^2$, what is $f(-3)$? Student's answer: -9

**Square the input.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Squaring a negative number gives a positive result. $(-3)^2 = 9$.** | **9** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the domain of $f(x) = 1/(x - 2)$? Student's answer: all real numbers

**Find any values that make the denominator zero.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **When x = 2, the denominator is 0. The domain is all real numbers except x = 2.** | **All real numbers except 2** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the range of $y = x^2$ when x is any real number? Student's answer: all real numbers

**Think about the smallest output the squares can give.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Squares are never negative, so the smallest output is 0. The range is $y \geq 0$.** | **$y \geq 0$** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $g(x) = 5 - x$, what is $g(2)$? Student's answer: 7

**Substitute 2 for x.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added 2 instead of subtracting. $g(2) = 5 - 2 = 3$.** | **3** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sequence is defined by $a(n) = 3n + 1$. What is $a(4)$? Student's answer: 12

**Substitute 4 for n.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student multiplied by 3 but did not add 1. $a(4) = 3(4) + 1 = 13$.** | **13** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A recursive sequence has $a(1) = 2$ and $a(n) = a(n - 1) + 3$. What is $a(3)$? Student's answer: 9

**Find each term from the one before it.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **$a(2) = 5$ and $a(3) = 8$. The student added 3 to the wrong term.** | **8** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = x^2 - 1$, what is $f(3)$? Student's answer: 10

**Substitute 3 and simplify.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added 1 instead of subtracting it. f(3) = 9 - 1 = 8.** | **8** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table gives f(0) = 5, f(1) = 7, and f(2) = 9. What is f(0)? Student's answer: 7

**Read the value from the table at x = 0.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **7 is the value at x = 1. At x = 0, f(0) = 5.** | **5** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = 2$, what is $f(3)$? Student's answer: 6

**Raise 2 to the power of 3.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student multiplied 2 by 3. $2^3 = 8$.** | **8** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sequence is 2, 6, 18, 54. What is the next term? Student's answer: 108

**Find the rule that gets from one term to the next.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Each term is 3 times the one before it. The next term is 54 × 3 = 162. The student doubled.** | **162** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the y-intercept of y = 3x - 6? Student's answer: 6

**Find y when x = 0.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The y-intercept is where x = 0, so y = -6. The student dropped the sign.** | **-6** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the x-intercept of y = 2x - 8? Student's answer: -8

**Set y = 0 and solve for x.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student used the slope, -1, instead of the y-intercept. When x = 0, y = 4.** | **4** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the maximum value of $y = -x^2 + 4$? Student's answer: 4 is the minimum.

**Decide whether the parabola opens up or down.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The negative coefficient means the parabola opens down, so the vertex at (0, 4) is the maximum.** | **4 is the maximum** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Over what interval is y = 2 increasing? Student's answer: it is decreasing everywhere

**Look at the direction of the graph.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Since 2 is greater than 1, the function increases for all real numbers.** | **All real numbers** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What are the zeros of $f(x) = x^2 - 9$? Student's answer: $x = 3$

**Set f(x) = 0 and solve.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student found only one zero. $x^2 = 9$ gives $x = 3$ or $x = -3$.** | **$x = 3$ and $x = -3$** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line passes through (0, 4) and (2, 0). What is its rate of change? Student's answer: 2

**Use (change in y) ÷ (change in x).**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The y-values go down, so the rate is negative. (0 - 4) ÷ (2 - 0) = -2.** | **-2** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s 100 gallons and drains 4 gallons per minute. Write f(t) for the amount after t minutes. Student writes: f(t)

**Decide whether the amount goes up or down.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The tank drains, so the amount decreases. f(t) = 100 - 4t.** | **f(t) = 100 - 4t** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ccount starts with $20 and gets $5 each week. Write f(w) for the amount after w weeks. Student writes: f(w

**Identify the starting amount and the weekly change.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The starting amount is $20, and the weekly gain is $5. f(w) = 20 + 5w.** | **f(w) = 20 + 5w** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sequence is 3, 6, 12, 24. Write an explicit formula for the nth term. Student writes: a(n) = 3n

**Check the ratio between terms.**

What did the student do wrong? Why do you think the student made this error?

**The terms double each time, so the formula is exponential. a(n) = 3 · 2^(n-1).**

Solve the problem correctly. Show all of your work.

**a(n) = 3 · 2^(n-1)**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = x + 2$ and $g(x) = 3x$, what is $f(g(2))$? Student's answer: 12

**Find g(2) first, then apply f.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student applied g to f(2). g(2) = 6, and f(6) = 8.** | **8** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

worker earns $15 per hour plus a fixed $30 bonus. Write the function for h hours. Student writes: f(h) = 30h +

**Decide which part depends on hours.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The hourly pay is multiplied by hours and the bonus is fixed. f(h) = 15h + 30.** | **f(h) = 15h + 30** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The formula F = (9/5)C + 32 converts Celsius to Fahrenheit. What is F when C = 20? Student's answer: 52

**Substitute and multiply first.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student added 20 and 32. (9/5)(20) = 36, and 36 + 32 = 68.** | **68** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has x = 0, 1, 2 and y = 2, 4, 8. Is it linear or exponential? Student's answer: linear

**Check whether the outputs have a constant difference or a constant ratio.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The differences are 2 and 4, but the ratio is 2 each time. The model is exponential.** | **Exponential** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = 3, 7, 11 for x = 0, 1, 2. Is it linear or exponential? Student's answer: exponential

**Check the differences.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The difference is 4 each time, so the model is linear.** | **Linear** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = 5, 10, 20 for x = 0, 1, 2. Is it linear or exponential? Student's answer: linear

**Check the ratio between outputs.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The ratio is 2 each time, not a constant difference. The model is exponential.** | **Exponential** |

Original task. Format inspired by error analysis routines.

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# Find the Error

F.LE.1

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A population increases by 10% each year. Is the model linear or exponential? Student's answer: linear

**Decide whether the change is a fixed amount or a percent.**

What did the student do wrong? Why do you think the student made this error?

**A percent change is multiplicative, so the model is exponential.**

Solve the problem correctly. Show all of your work.

**Exponential**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ccount starts with $500 and gains $20 each month. Is the model linear or exponential? Student's answer: e

**Decide whether the change is a fixed amount or a percent.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The account gains the same amount each month, so the model is linear.** | **Linear** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = 2, 6, 18 for x = 0, 1, 2. Is it linear or exponential? Student's answer: linear

**Check the ratio and the difference.**

What did the student do wrong? Why do you think the student made this error?

**The ratio is 3 each time, and the differences change. The model is exponential.**

Solve the problem correctly. Show all of your work.

**Exponential**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write a linear function through (1, 5) and (3, 9). Student writes: $y = 5x + 1$

**Find the slope, then the y-intercept.**

What did the student do wrong? Why do you think the student made this error?

**The slope is (9 - 5) ÷ (3 - 1) = 2. Then 5 = 2(1) + b, so b = 3. y = 2x + 3.**

Solve the problem correctly. Show all of your work.

**$y = 2x + 3$**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A population of 100 grows by 50% each year. Write the exponential model. Student writes: y = 100(0.5)^t

**Decide whether the growth is up or down.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **50% growth means the factor is 1 + 0.5 = 1.5. The model is y = 100(1.5)^t.** | **y = 100(1.5)^t** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

tial function passes through (0, 3) and doubles each time x increases by 1. Write the function. Student write

**Start with the value at x = 0, then multiply by the growth factor.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The starting value is 3 and the growth factor is 2. The function is y = 3 · 2^x.** | **y = 3 · 2^x** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A tank starts with 40 gallons and loses 4 gallons per year. Write a linear model. Student writes: $y = 40 + 4t$

**Decide whether the amount increases or decreases.**

What did the student do wrong? Why do you think the student made this error?

**The amount decreases by 4 each year.**
**The model is $y = 40 - 4t$.**

Solve the problem correctly. Show all of your work.

**$y = 40 - 4t$**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A value of 500 decreases by 8% each year. Write the exponential model. Student writes: y = 500(1.08)^t

**Decide whether the amount grows or decays.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Decreasing by 8% leaves 92% each year. The model is y = 500(0.92)^t.** | **y = 500(0.92)^t** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line has slope -3 and passes through (2, 1). Write the equation. Student writes: y = 3x + 7

**Use y = mx + b and the point to find b.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The slope is -3, not 3. 1 = -3(2) + b gives b = 7, so y = -3x + 7.** | **y = −3x + 7** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the median of the data 3, 5, 5, 7, 9. Student's answer: 6

**Find the middle value of the ordered data.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The data is already in order, and the middle value is 5. The student averaged the wrong numbers.** | **5** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which display is best for categorical data, such as favorite colors? Student's answer: histogram

**Think about what a histogram shows.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Histograms show numerical data in intervals. A bar chart is best for categories.** | **Bar chart** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box plot has Q1 = 3 and Q3 = 9. What is the interquartile range? Student's answer: 12

**Subtract Q1 from Q3.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added the values. IQR = Q3 - Q1 = 9 - 3 = 6.** | **6** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A histogram has bars with frequencies 2, 5, and 3. How many data values are there? Student's answer: 8

**Add all the frequencies.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added the wrong numbers. 2 + 5 + 3 = 10.** | **10** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a stem-and-leaf plot, 4 | 1 means what value? Student's answer: 4.1

**Combine the stem and the leaf.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The stem is the tens digit and the leaf is the ones digit. 4 \| 1 means 41.** | **41** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lata set has minimum 2, Q1 = 4, median = 6, Q3 = 8, and maximum 14. What is the range? Student's answer

**Subtract the minimum from the maximum.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **14 is the maximum, not the range. The range is 14 - 2 = 12.** | **12** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

andard deviation of 2. Group B has a mean of 55 and a standard deviation of 2. Which group has the highe

**Compare the means.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The mean of Group B is 55, which is higher than 50. Group B has the higher center.** | **Group B** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

andard deviation of 1 and Group B has a standard deviation of 5. Which group has more spread? Student's

**Compare the standard deviations.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A larger standard deviation means more spread. Group B has more spread.** | **Group B** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

a set has an outlier of 100. Which center is less affected by the outlier: mean or median? Student's answer: r

**Think about how an outlier changes a total.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The mean is pulled toward the outlier. The median is less affected because it is the middle value.** | **Median** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

value of 80 is in a data set with mean 70 and standard deviation 5. What is its z-score? Student's answer: 1

**Use z = (value − mean) ÷ standard deviation.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student did not divide by the standard deviation. z = (80 − 70) ÷ 5 = 2.** | **2** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

eviation of 3. Group B has a mean of 14 and a standard deviation of 3. How many standard deviations is B's

**Divide the difference in means by the standard deviation.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The difference is 2, and 2 ÷ 3 = 2/3 of a standard deviation.** | **2/3** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pproximately normal. About what percent of values lie within 2 standard deviations of the mean? Student'

**Recall the empirical rule.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **About 68% lies within 1 standard deviation. About 95% lies within 2 standard deviations.** | **About 95%** |

Original task. Format inspired by error analysis routines.